Shared-subspace near–far selection

01 Shared view

Post-RoPE Q / K

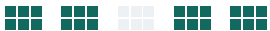
128D → 32D

16 complete pairs

Original Q/K/V

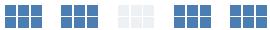
02 Regional L1

Far: extrema



96 pages

Near: mean key



16 pages

03 Token L2

Full-history scoring

Mask → per-Q softmax

→ GQA probability mean

Single-pool TopK

04 Original attention

Indices protection

Gather original K/V

Original Q · 128D

Per-Q attention

Original Q/K/V bypass · protected tokens retained

Optional far gate: prefill → freeze → reuse; OFF skips far L1 + L2

Disabled by default